BODAPATI VISHNU SAI VARDHAN

Software Engineer | Full-Stack, Systems & Real-Time Graphics Developer

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EDUCATION

Duke University — M.Eng, Game Design, Development and Innovation *Aug 2025 – Present*

Pratt School of Engineering | Coursework: Advanced Game Design, Unreal Engine Development, C++ Programming

University of Wollongong — B.S. Computer Science, Game and Mobile Development *Sep 2021 – Sep 2024*

Coursework: Game Engine Essentials, Interactive Computer Graphics, Mobile Application Development, Software Development Methodologies

TECHNICAL SKILLS

Languages: C#, C++, C, Python, JavaScript, TypeScript, Java, Lua, SQL

Web & Cloud: React, Next.js, Node.js, Tailwind CSS, REST APIs, Docker, Google Cloud Run, Gemini API

Systems & Tools: Unity, Unreal Engine, Git/GitHub, real-time rendering, performance profiling, XR/OpenXR, CI-style build & QA workflows

EXPERIENCE

Research Assistant — Duke Intelligent Interactive Internet of Things (I3T) Lab *Oct 2025 – Present*

- Build and iterate on real-time VR software systems for a clinical rehabilitation application, optimizing frame-rate stability, input responsiveness, and interaction reliability under strict performance constraints.
- Translate clinical requirements into engineering specs; design testable, modular interaction systems used across repeated research sessions.
- Evaluate integration paths for 3D Gaussian Splatting and AI-generated content pipelines to extend the platform's rendering capabilities.

PROJECTS

Draft USA — Full-Stack AI Web Platform *May 2026*

Next.js 14, TypeScript, Tailwind CSS, Gemini API, Docker, Google Cloud Run, Recharts

- Designed and shipped a full-stack web app end to end, including server-side API routes integrating the Gemini API for recommendation, analysis, and summarization features.
- Containerized and deployed the application on Google Cloud Run; built data-driven dashboard views with Recharts.

MR Blueprint — Mixed Reality Systems Sandbox *Jan – Apr 2026*

Unity, C#, OpenXR, Meta Quest 3/3S, XR Interaction Toolkit

- Engineered core C# systems for object spawning, selection, transform manipulation, and a real-time physics simulation pipeline with adjustable mass, friction, and gravity parameters.
- Built a live data-graphing and snapshot/restore tool (PhysicsLens) for rapid experimentation without manual scene rebuilds.

CinemaScout — Real-Time 3D Spatial Reconstruction Tool *Jul 2026*

Unity, C#, 3D Gaussian Splatting, OpenXR, WebSpatial

- Built a real-time Unity renderer for 3D Gaussian Splat reconstructions, plus a configurable virtual camera system (focal length, FOV, spline-based motion) with live in-headset preview.

Additional Projects: Tower of Tricks (Unreal Engine UI/build systems), Hungry Owl (sole developer, Playdate/Lua), Meteor Mayhem (Python computer-vision game controls), Overpriced & The Misadventures of Skylar Knight (Unreal Engine game jams).

AWARDS

- 1st Place – Grand Winner, DesignXR Hackathon 2026 — MR Blueprint
- 1st Place: Best Spatial Reconstruction Project, 1st Place: Best App Running on Emulator & Runner-Up: Best Interactive World Experience, Worlds in Action Hack [02-LA] — CinemaScout
- Winner – Honorable Mention, Team USA × Google Cloud Hackathon — Draft USA
- Top 50 Semifinalist of 1,323 teams, DevStudio 2026 by Logitech — MR Blueprint